

BANK LOAN ANALYSIS REPORT

Power BI · SQL · Data Analytics

38,576 Total Applications Loan records analysed	\$435.8M Total Funded Disbursed amount	\$473.1M Amount Received Repayments collected	86.2% Good Loans Portfolio health
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Data Analyst Portfolio Project

MS SQL Server · Power BI Desktop · June 2023

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1. Project Overview

This project delivers a comprehensive end-to-end bank loan portfolio analysis using **MS SQL Server** for data extraction and business queries, and **Power BI Desktop** for interactive dashboard visualisation. The dataset covers 38,576 loan records with a combined funded value of \$435.8M, enabling analysis of portfolio health, borrower profiles, repayment trends, and risk segmentation.

Objectives

- Track core KPIs — Total Applications, Funded Amount, Amount Received, Avg Interest Rate, Avg DTI
- Segment the portfolio into Good Loans vs Bad Loans for risk assessment
- Analyse monthly trends, loan term distribution, employment history, and loan purpose
- Provide a borrower-level drill-through details view for granular analysis
- Validate all Power BI metrics against SQL query results for accuracy

Tools & Technology Stack

Tool	Version	Purpose
MS SQL Server	19.0	Data storage, querying, business logic
SQL Server Mgmt Studio	19.0.20209.0	Query writing, database management
Power BI Desktop	June 2023	Dashboard design, DAX, visualisations
MS Excel	2021	Initial data inspection and validation

2. Dashboard 1 — Summary

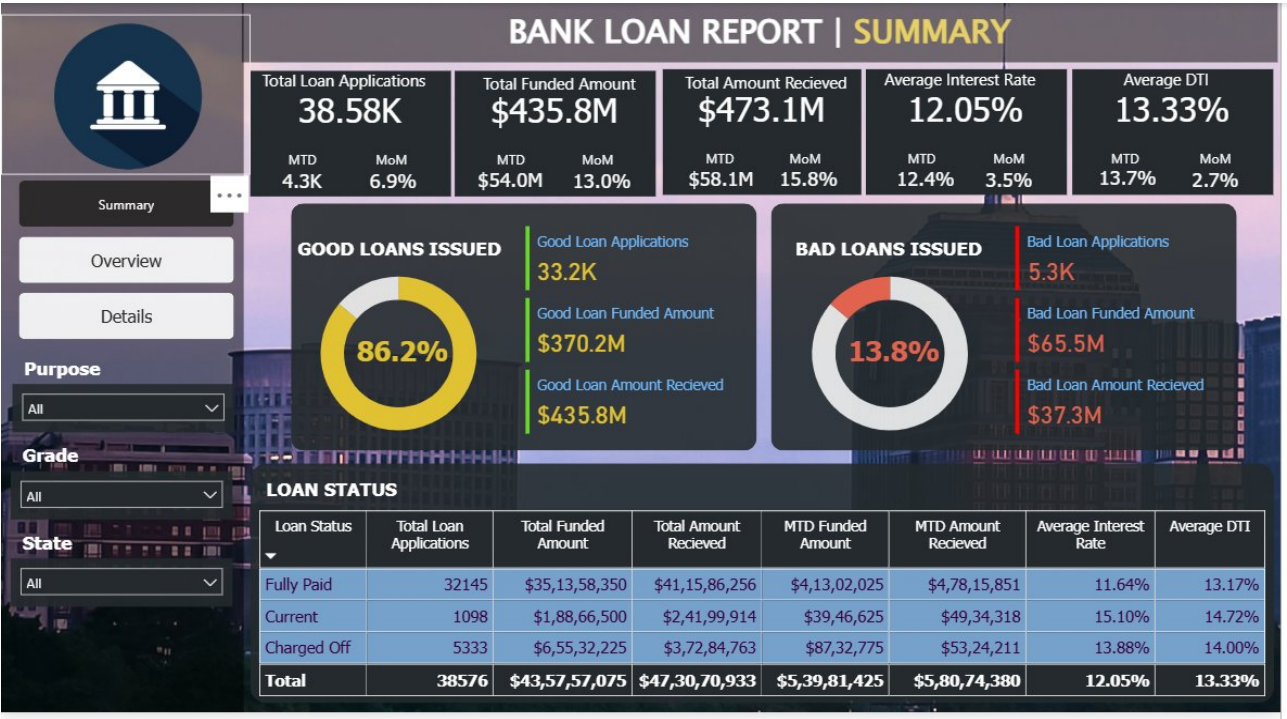


Fig 1. Summary Dashboard — KPI Cards, Good/Bad Loan Segmentation, Loan Status Table

Core KPIs

38,576 Total Applications MTD: 4.3K MoM: +6.9%	\$435.8M Total Funded MTD: \$54.0M MoM: +13.0%	\$473.1M Amt Received MTD: \$58.1M MoM: +15.8%	12.05% Avg Interest Rate MTD: 12.4% MoM: +3.5%	13.33% Avg DTI MTD: 13.7% MoM: +2.7%
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Good Loans vs Bad Loans

Metric	✓ Good Loans (86.2%)	✗ Bad Loans (13.8%)
Applications	33,200	5,300
Funded Amount	\$370.2M	\$65.5M
Amount Received	\$435.8M	\$37.3M

Loan Status Breakdown

Loan Status	Applications	Funded Amount	Amt Received	Avg Interest	Avg DTI
Fully Paid	32,145	\$35.1M	\$41.2M	11.64%	13.17%
Current	1,098	\$1.9M	\$2.4M	15.10%	14.72%

Charged Off	5,333	\$6.6M	\$3.7M	13.88%	14.00%
Total	38,576	\$43.6M	\$47.3M	12.05%	13.33%

3. Dashboard 2 — Overview

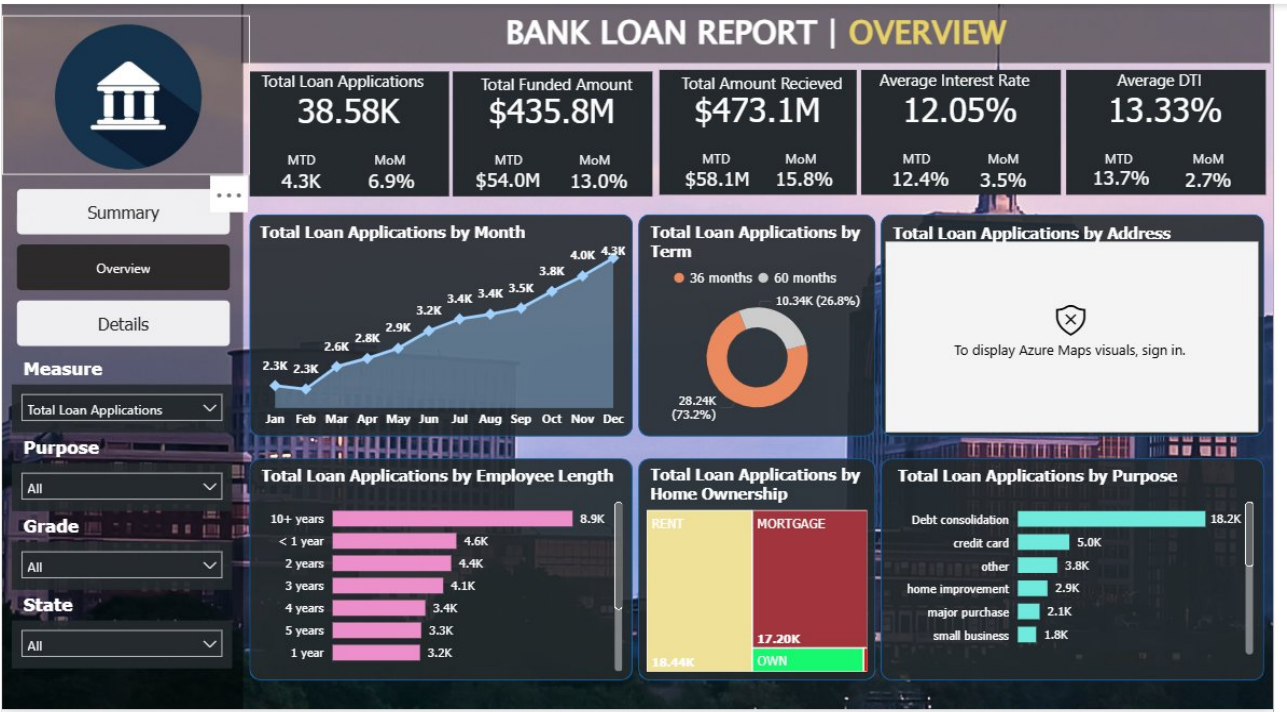


Fig 2. Overview Dashboard — Monthly Trend, Loan Term, Employee Length, Home Ownership, Purpose

The Overview dashboard provides a multi-dimensional view of lending activity through six visual components, enabling trend analysis, borrower profiling, and product distribution insights.

Visual	Chart Type	Key Insight
Monthly Applications Trend	Line Chart	Steady growth Jan (2.3K) → Dec (4.3K)
Loan Term Distribution	Donut Chart	73.2% on 36-month; 26.8% on 60-month
Employment Length	Bar Chart	10+ years highest at 8.9K applications
Home Ownership	Tree Map	Mortgage (17.2K) > Rent (16.4K) > Own
Loan Purpose	Bar Chart	Debt consolidation leads at 18.2K (47%)
Geographic Distribution	Azure Map	Regional lending concentration by state

Monthly Applications — Detailed View

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2.3K	2.3K	2.6K	2.8K	2.9K	3.2K	3.4K	3.4K	3.5K	3.8K	4.0K	4.3K

4. Dashboard 3 — Details

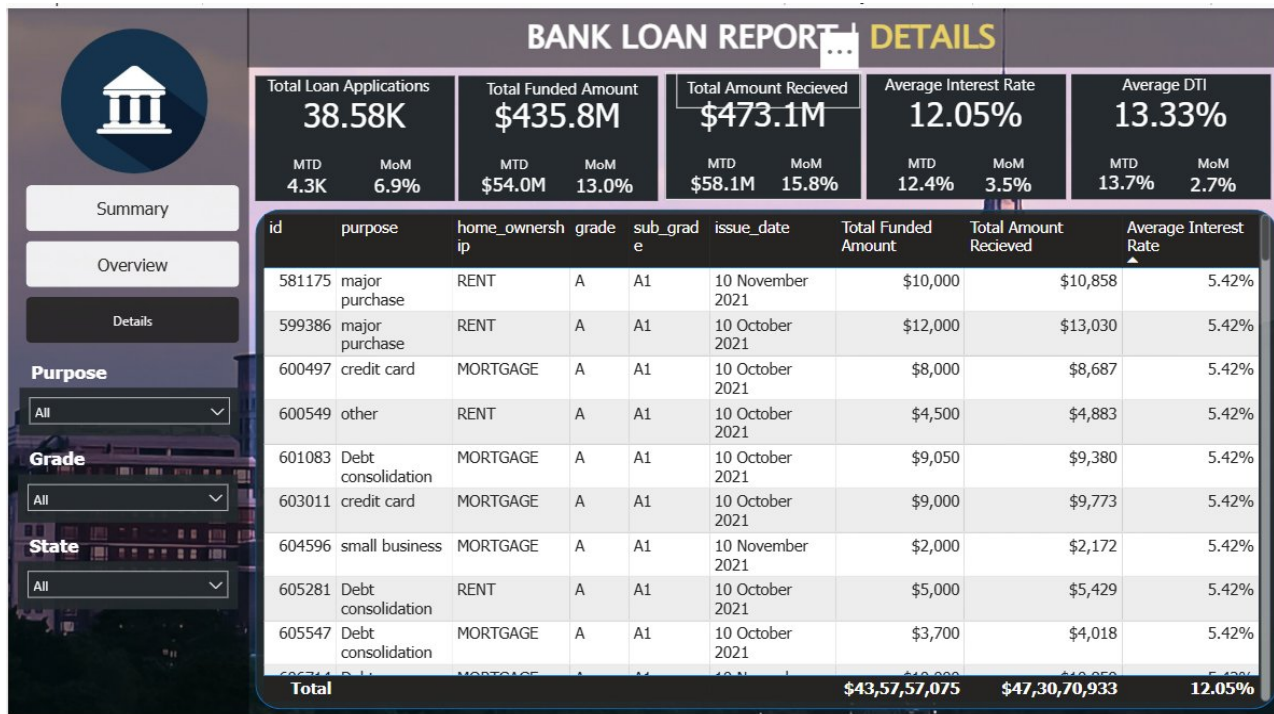


Fig 3. Details Dashboard — Row-level drill-through table with loan-level data

The Details dashboard provides a borrower-level consolidated table enabling granular investigation of individual loans. Users can filter by Purpose, Grade, and State to drill into specific segments.

Columns Available in Details View

Column	Description
ID	Unique loan identifier
Purpose	Reason for loan (debt consolidation, credit card, etc.)
Home Ownership	Borrower's housing status (RENT / MORTGAGE / OWN)
Grade / Sub-Grade	Credit quality tier assigned by the bank
Issue Date	Date the loan was originated
Total Funded Amt	Disbursed principal amount
Total Amt Received	Cumulative repayments collected
Avg Interest Rate	Annual rate charged on the loan

Sample Records — Grade A1 Loans

Loan ID	Purpose	Ownership	Issue Date	Funded	Received	Rate
581175	Major Purchase	RENT	10 Nov 2021	\$10,000	\$10,858	5.42%
599386	Major Purchase	RENT	10 Oct 2021	\$12,000	\$13,030	5.42%
600497	Credit Card	MORTGAGE	10 Oct 2021	\$8,000	\$8,687	5.42%
601083	Debt Consol.	MORTGAGE	10 Oct 2021	\$9,050	\$9,380	5.42%
604596	Small Business	MORTGAGE	10 Nov 2021	\$2,000	\$2,172	5.42%

5. Key Findings & Business Insights

86.2% Healthy Portfolio Fully Paid + Current loans constitute 86.2% — the bank's core portfolio is low-risk and performing well.	47% Debt Consolidation Nearly half all applications are for debt consolidation, making it the bank's single largest loan product.
+86% Year-on-Year Growth Applications grew from 2.3K in January to 4.3K in December — an 86% increase across the year.	73.2% Short-Term Preference Borrowers strongly prefer 36-month terms (73.2%) over 60-month, suggesting demand for quicker loan closure.
8.9K Experienced Borrowers Applicants with 10+ years of employment dominate at 8.9K — the bank's most creditworthy segment.	13.8% Charge-Off Risk 5,333 Charged-Off loans worth \$6.6M funded — only \$3.7M recovered, highlighting a collection gap.
+15.8% MoM Recovery Growth Amount Received grew 15.8% MoM, outpacing funded amount growth of 13% — strong repayment momentum.	

Risk Observation

Charged-Off loans have an average interest rate of 13.88% and DTI of 14.00% — both above portfolio averages. This suggests that higher-risk borrowers (higher DTI, higher rate) are defaulting at elevated rates and the bank may benefit from tightening underwriting criteria for DTI > 14%.

6. SQL Analysis Approach

All Power BI KPIs and metrics were independently validated using MS SQL Server queries. SQL results were compared against Power BI visuals to ensure data accuracy across both tools.

Business Problem	SQL Approach
Total loan applications & MTD count	COUNT(*) with MONTH() / DATEPART() filter
Total funded & received amounts	SUM() on funded_amount and amount_received
Month-over-Month % changes	CTE with LAG() or PARTITION BY for prior month
Good vs Bad loan segmentation	CASE WHEN loan_status IN ('Fully Paid','Current')
Average Interest Rate & DTI by status	AVG() grouped by loan_status
Monthly trend (applications by month)	GROUP BY DATENAME(MONTH, issue_date) ORDER BY MONTH()
Loan purpose breakdown	GROUP BY purpose ORDER BY COUNT(*) DESC
Employment length distribution	GROUP BY emp_length ORDER BY COUNT(*) DESC

Key SQL Functions Used

- Aggregation:** COUNT(), SUM(), AVG(), DISTINCT
- Date Functions:** DATEPART(), DATENAME(), MONTH(), CAST(), DAY(), QUARTER()
- Filtering:** WHERE, HAVING, GROUP BY, ORDER BY
- Advanced:** CTE (WITH clause), PARTITION BY, window functions
- Data Types:** DECIMAL(), type casting for percentage calculations

7. Power BI Skills Demonstrated

Category	Skills / Features
Data Connection	Connect to MS SQL Server; Power Query for data cleaning
Data Modelling	Relationships, date tables, calculated columns
DAX Measures	Time Intelligence (MTD, MoM%), SUM, CALCULATE, FILTER, DIVIDE
Visualisations	KPI cards, donut charts, line charts, bar charts, tree maps, Azure Maps
UX / Navigation	Page navigation buttons, bookmarks, slicers (Purpose/Grade/State)
Formatting	Dark theme, conditional formatting, custom colour palette

8. Conclusion

This Bank Loan Analysis project demonstrates a complete data analytics workflow — from raw SQL querying through to interactive Power BI dashboards — applied to a real-world BFSI (Banking, Financial Services & Insurance) dataset. The three-page dashboard architecture effectively addresses three distinct analytical needs: executive KPI summary, operational trend overview, and borrower-level detail.

Project Strengths

- ✓ End-to-end workflow: SQL data extraction validated against Power BI metrics
- ✓ BFSI domain relevance: Loan portfolio, MTD/MoM tracking, and risk segmentation mirror real NBFC/bank reporting
- ✓ Clear stakeholder communication: Executive summary, operational drill-down, and row-level detail across 3 pages
- ✓ Strong DAX implementation: Time Intelligence functions, Good/Bad loan classification, dynamic filters
- ✓ Business impact framing: Findings linked to actionable insights (underwriting risk, product demand, borrower profile)

Potential Enhancements

- Add a drill-through from the Summary donut to Details for Good/Bad loan breakdown
- Fix and integrate the Azure Maps geographic visual for regional analysis
- Introduce a Tooltip page showing mini-trend charts on hover
- Add predictive scoring: flag high-DTI + high-rate applicants as charge-off risk

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